The Value of an Active Contact: Payoff and Illustrative Thresholds

1. Adding a contact value to the one-day game

Consider a person choosing between attending (A) and staying home (H). Nobody knows anyone's initial infection status, including their own. Each person is independently infected with probability π. If k other people attend and are contacts of the person, each is both infected and able to transmit across the contact with probability $\pi\beta$. Thus, conditional on the person being initially susceptible, the probability of a new infection is

$$q(k) = 1 - (1 - \pi\beta)^k. \tag{1}$$

The extra expected infection cost of attending, relative to staying home, is

$$d(k) = L(1 - \pi)q(k) = L(1 - \pi)\big[1 - (1 - \pi\beta)^k\big]. \tag{2}$$

In the original one-day game, staying home costs c, so the gain from attending rather than staying home is $c - d(k)$. To give each participant a benefit v from each active contact, an attendee with k attending contacts receives vk. The extended expected payoffs are

$$U(A \mid k) = -L\big[\pi + (1 - \pi)q(k)\big] + vk, \tag{3}$$

$$U(H \mid k) = -L\pi - c. \tag{4}$$

Subtracting gives

$$\boxed{U(A \mid k) - U(H \mid k) = c + vk - d(k).} \tag{5}$$

Here v is a *per-person, per-contact, per-day* benefit. An active edge joins two attendees and therefore contributes $2v$ to their combined payoff. Under this interpretation, c should represent the cost of staying home *apart from* the contact benefits already counted by v.

2. An illustrative welfare break-even value

The community-network experiment reports mean active contact days C_s and mean currently infected people M_s over days 1–84 for each schedule s. For this calculation only, suppose that an infected person incurs cost L *on each day they are infected*. If H_s is the number of person-days spent at home, the expected 84-day group payoff, up to common terms, is

$$W_s(v) = 2vC_s - cH_s - L(84)M_s. \tag{6}$$

All compared rotation splits have the same total attendance, so their H_s terms cancel in a welfare difference.

At $\beta = 0.085$, the saved daily-switching results give

$$C_{\text{whole}} = 28{,}612.5, \qquad C_{\text{even}} = 16{,}020.9,$$
$$M_{\text{whole}} = 15.49, \qquad M_{\text{even}} = 1.80.$$

The whole-community split therefore adds

$$\Delta C = 28{,}612.5 - 16{,}020.9 = 12{,}591.6 \quad \text{active edge-days,} \tag{7}$$
$$84\Delta M = 84(15.49 - 1.80) = 1{,}149.96 \quad \text{infected person-days.} \tag{8}$$

With the illustrative value $L = 10$, its additional contact benefit is $2v(12{,}591.6)$ and its additional infection cost is 11,499.6. Equating the two yields

$$v_{\text{BE}} = \frac{10 \cdot 84 \cdot (15.49 - 1.80)}{2(28{,}612.5 - 16{,}020.9)} = \frac{11{,}499.6}{25{,}183.2} \approx 0.457. \tag{9}$$

This is a break-even point *under the stated payoff assumptions*, not an empirically measured contact value. If L is instead charged only when a new infection occurs, equation (6) and the break-even value must be recalculated from incident infections.

3. An illustrative obedience threshold in the three-person game

In the three-person complete graph, consider the fair rotation over profiles with exactly two attendees. The person recommended to stay home faces $k = 2$ attending contacts if they deviate to A. At $\pi = 0.1$, $\beta = 0.04$, $L = 10$, and $c = 0.05$, equation (2) gives

$$d(2) = 10(0.9)\left[1 - (1 - 0.1 \cdot 0.04)^2\right] \tag{10}$$
$$= 9(1 - 0.996^2) = 0.071856. \tag{11}$$

The home recommendation is obeyed only if

$$U(H \mid 2) \geq U(A \mid 2) \quad \Longleftrightarrow \quad c + 2v \leq d(2). \tag{12}$$

Consequently,

$$\boxed{v \leq \frac{0.071856 - 0.05}{2} = 0.010928.} \tag{13}$$

The attendee's recommendation is also obeyed for every $v \geq 0$, because $d(1) = 0.036$ and $c + v \geq 0.05 > 0.036$. Thus, within this toy game, the fair two-attendee rotation remains an equilibrium for $0 \leq v \leq 0.010928$.

Equations (9) and (13) answer different questions. The first compares aggregate welfare across fixed 84-day schedules at $\beta = 0.085$; the second tests an individual's incentive to obey a one-day recommendation at $\beta = 0.04$. They are illustrative thresholds, not two estimates of the same empirical quantity.